

Thiago Vasco

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EDUCATION

University of British Columbia

Present

Master of Science, Mathematics, Institute of Applied Mathematics (IAM), Cumulative Average 88.6%

- Co-supervised by Prof. Philip Loewen and Prof. Bhushan Gopaluni
- Writing a thesis on how to make LP-MPC controllers more interpretable in the context of large industrial processes; doing research in tandem with the Burnaby Refinery in British Columbia, Canada

University of British Columbia

Graduated: 12/23

Bachelor of Science, Major in Statistics, Minor in Mathematics, Cumulative Average 85.3%

- UBC Centennial Scholars Entrance Award, Dean's Honour List, British Columbia Government Scholarship, Professor Richard Lipsey Award in International Economics

EMPLOYMENT

Intern Analyst, Applied Research & Models, CPP Investments, Toronto ON

05/22 – 08/22

- Developed knowledge of professional software engineering practices for a full software development cycle, including continuous integration, coding standards, and code reviews
- Added functionality to analyze factor and cash exposures within asset allocation software; helped research novel factor investing methodology which better characterizes idiosyncratic risk of illiquid asset classes
- Handled other ad hoc requests such as data cleaning and updating software packages

Intern Analyst, Portfolio Design & Construction, CPP Investments, Toronto ON

01/22 – 04/22

- Debugged and added functionality to experimental dynamic portfolio optimization model created by the team
- Modified and ran regressions for risk premia forecasting and parameter estimation; also conducted stationarity testing
- Created a high-quality GUI using R Shiny for researching optimal factor exposures under different model assumptions

Project Assistant, CANARY, UBC Vancouver Stroke Program, Vancouver BC

05/21 – 08/21

- Developed software for training and testing of Hugging Face transformer models applied to NLP classification; contrasted performance with simpler models and other types of neural architectures such as Hierarchical Attention Networks
- Wrote code to parse language data and to accomplish other preprocessing tasks

TECHNICAL PROJECTS

LQR Missile Guidance

- Implemented an LQR-based control strategy for controlling a nonlinear, time-varying system using Python; this was done as a class project for MECH 509

Primal-Dual Interior-Point Methods with Application to Optimal Control

- Implemented two primal-dual interior-point methods using Julia for solving linear and nonlinear problems in optimal control; this was done as a class project for CPSC 536M

Model-Based Reinforcement Learning Using Predictive Sampling MPC

- Implemented a model-based RL algorithm as a class project for CPSC 533V; system dynamics are modeled using a neural network, and optimal actions are found by using a sampling-based MPC approach

ADDITIONAL INFORMATION

- **Legal Status:** Canadian Citizen
- **Academic Interests:** Control, Optimization, Reinforcement Learning, Machine Learning
- **Languages & Frameworks:** Python, PyTorch, Java, C, C++, R, Julia, MATLAB, GitHub, Jira, Confluence, LATEX, and others
- **Relevant Courses:** Calculus I-IV, Linear Algebra, Probability, Statistics, Machine Learning, Time Series, Differential Equations, Control Theory, Numerical Optimization, OOP Software Engineering, Algorithms